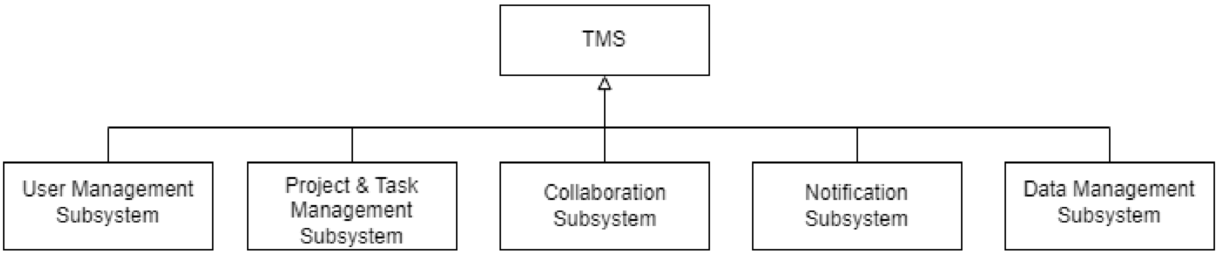
Subsystem Decomposition

This section introduces the Task Management System decomposition into subsystems. Based on the functional requirements, the following subsystems were identified: **User management**, **project and task management**, **collaboration**, **notification**, and **data management**.

Figure 1

The UML Structure Diagram of the TMS Architecture



Source: Own results.

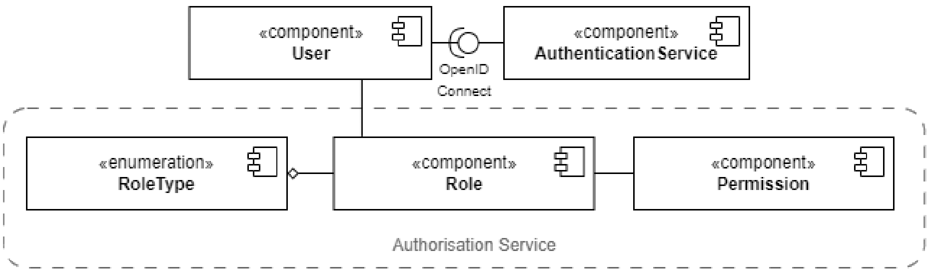
1. User Management Subsystem

The user management subsystem manages user profiles and user personal information. It also controls user authentication and user access to different parts of the TMS by implementing role-based access control (RBAC).

The user management subsystem contains the following components: `User`, Authorisation Service, and `AuthenticationService`. Authentication Service is a collection of three components: `Permission`, `Role`, and `RoleType`. The `User` and `AuthenticationService` are connected via the OpenID Connect interface.

Figure 2

The UML Component Diagram of the TMS User Management Subsystem



Source: Own results.

2. Project and Task Management Subsystem

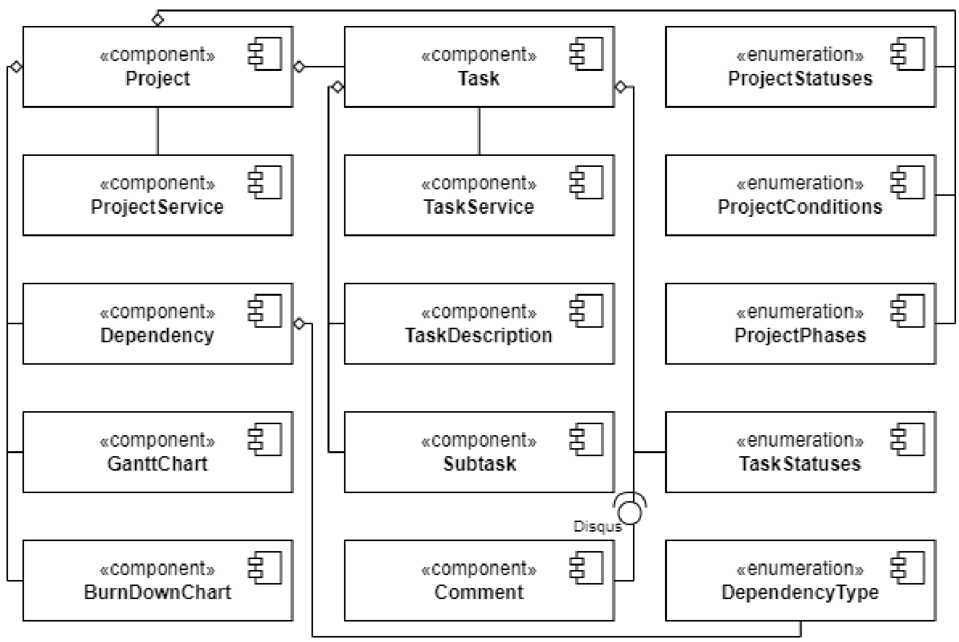
The project and task management subsystem allows the user to execute actions concerning a project: the project CRUD operations, arrange task order, organise task dependencies, track project status, condition, and phase, monitor project risks, issues, and damages, track project progress, read the Gantt and Burn-down charts, and generate project reports.

The project and task management subsystem also allows the user to execute actions concerning a task: the task and subtask CRUD operations, assign tasks, tag tasks, arrange subtask order, track subtask status, comment tasks, and generate task reports.

The project and task management subsystem contains the following components: `Project`, `Task`, `ProjectService`, `TaskService`, `Dependency`, `DependencyType`, `GanttChart`, `BurnDownChart`, `TaskDescription`, `Subtask`, `Comment`, `ProjectStatuses`, `ProjectConditions`, `ProjectPhases`, and `TaskStatuses`.

Figure 3

The UML Component Diagram of the TMS Project and Task Management Subsystem



Source: Own results.

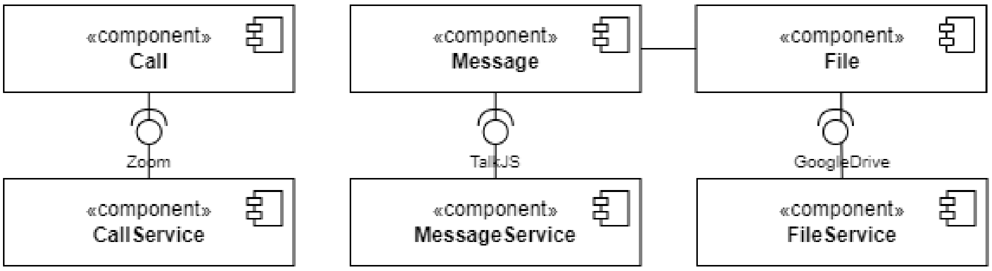
3. Collaboration Subsystem

The collaboration subsystem allows the user to send messages to a user or a group of users, receive messages, make audio and video calls, and share images, videos, and text files.

The collaboration subsystem contains the following components: message, file, call, message service, file service, and call service. The `Call` and `CallService` components use the Zoom interface to interact. The `Message` and `MessageService` components use the TalkJS interface to interact. The `File` and `FileService` components use the GoogleDrive interface to interact.

Figure 4

The UML Component Diagram of the TMS Collaboration Subsystem



Source: Own results.

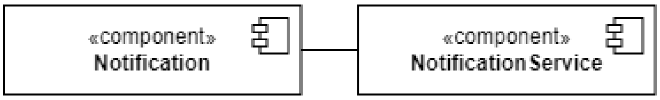
4. Notification Subsystem

The notification subsystem allows the user to execute the notification CRUD operations and send and receive notifications.

The notification subsystem contains the `Notification` and `NotificationService` components.

Figure 5

The UML Component Diagram of the TMS Notification Subsystem



Source: Own results.

5. Data Management Subsystem

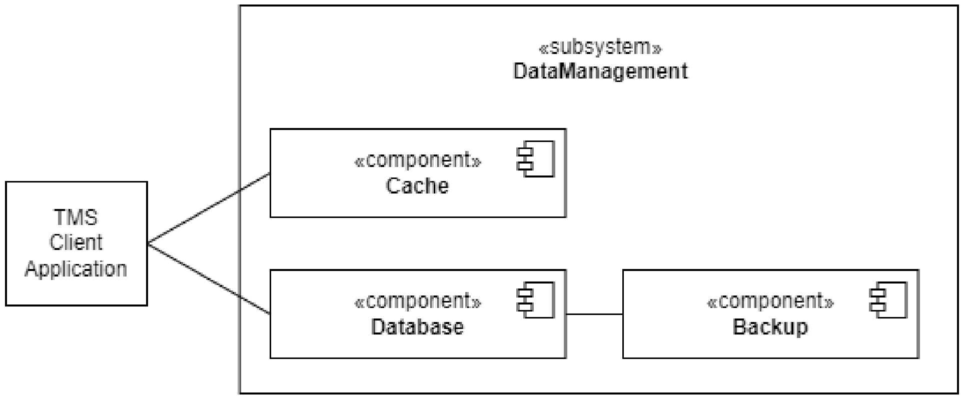
The data management subsystem manages the data storage and retrieval. It contains the following components: `Database`, `Cache`, and `Backup`.

The database component manages the storage and retrieval of data. The database interacts with the backup storage to backup data. The cache component is used for temporary data storage so that data requests are served faster. The backup component stores a copy of the original data to prevent data loss.

The client application requests data from the cache, the cache retrieves it quickly. In the case that the data is not stored in the cache, this request is sent to the database.

Figure 6

The UML Component Diagram of the TMS Data Management Subsystem



Source: Own results.